DVD Rental Business Performance Dashboard

Power BI & PostgreSQL DirectQuery Analysis

Junior Data Analyst Portfolio

Tools: Power BI, PostgreSQL, DirectQuery, DAX

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Project Overview

Objective: Analyze DVD rental business performance using Power BI connected to PostgreSQL through DirectQuery to generate real-time business insights.

Business Problem:

- Monitor revenue and rental activity
- Compare store performance
- Identify top-performing films
- Track monthly business trends
- Support data-driven decision making

Business Value:

- Real-time performance monitoring
- Better operational insights
- Improved business strategy
- Faster decision making

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Data Architecture

Data Source: PostgreSQL DVD Rental Database

Connection Mode: DirectQuery (Real-Time Data Connection)

Fact Tables:

- payment
- rental

Dimension Tables:

- film
- store
- inventory
- customer

Data Flow: Store → Inventory → Film Customer → Rental → Payment

Purpose: Efficient revenue and rental analysis using relational data modeling.

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Dashboard Overview

The dashboard provides a centralized business monitoring system.

Main Components:

01

KPI Section

02

Monthly Revenue Analysis

03

Monthly Rentals Analysis

04

Top Films Performance

05

Store Performance Table

06

Filter by Month and Store

Purpose: Provide both high-level and detailed business insights.

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Key Business Metrics

$30K

Total Revenue

7,211

Total Rentals

4.25

Average Revenue per Rental

759

Total Films

Insight: The business generates stable revenue with strong rental activity and a consistent pricing structure.

Business Meaning: The DVD rental operation shows steady performance and healthy customer engagement.

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Monthly Revenue Trend

February: around $5K

March: around $12K

April: around $14K

May: near zero

Insight: Revenue increased significantly from February to April.

Interpretation: Customer demand is growing steadily.

Possible Issue: May data may be incomplete or rental activity dropped.

Business Meaning: April represents peak performance.

Recommendation: Analyze marketing and operational strategy used in April and replicate it.

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Monthly Rental Trend

February: 1K rentals

March: 2.8K rentals

April: 3.3K rentals

May: near zero

Insight: Rental activity increases consistently over time.

Interpretation: Customer engagement is growing.

Business Meaning: Higher rental volume leads to higher revenue.

Recommendation: Maintain film availability and improve customer retention strategies.

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Store Performance Analysis

Store 1:

Revenue: $30,628

Rentals: 7,211

Store 2:

Revenue: $30,683

Rentals: 7,385

Insight: Store 2 slightly outperforms Store 1.

Interpretation: Store 2 is more productive and efficient.

Possible Reasons:

- Better location
- Higher customer traffic
- Better inventory
- Stronger operational strategy

Recommendation: Use Store 2 as a benchmark to improve Store 1 performance.

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Top Performing Films

Top 5 films generate the highest rentals.

Examples:

- Barbarella Streetcar
- Homeward Cider
- Horn Working
- Madness Attacks
- Movie Shakespeare

Insight: Customer demand is concentrated on specific films.

Business Meaning: Popular films drive most of the revenue.

Recommendation:

- Promote top films
- Increase stock availability
- Create featured movie sections
- Use targeted promotions

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Revenue and Rental Relationship

Observation: Revenue increases when rentals increase.

Insight

Rental volume directly impacts revenue.

Interpretation

The business model is stable and predictable.

Business Meaning

Increasing rental activity will directly increase revenue.

Recommendation: Focus on marketing and customer acquisition strategies.

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